MODEL COMPRESSION VIA COLLABORATIVE DATA-FREE KNOWLEDGE DISTILLATION FOR EDGE INTELLIGENCE

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ABSTRACT

Model compression without the original data for fine-tuning is challenging for deploying large-size models on resource constrained edge devices. To this end, we propose a novel data-free model compression framework based on knowledge distillation (KD), where multiple teachers are utilized in a collaborative manner to enable reliable distillation. It mainly consists of three components: adversarial data generation, multi-teacher KD, and adaptive outputs aggregation. In particular, some synthesized data are generated in an adversarial manner to mimic the original data for model compression. Then a multi-header module is developed to simultaneously leverage diverse knowledge from multiple teachers. The distillation outputs are adaptively aggregated for final prediction. The experimental results demonstrate that our framework outperforms the data-free counterpart significantly (4.48% on MNIST and 2.96% on CIFAR-10). Effectiveness of different components of our method is also verified via carefully designed ablation study.

***Index Terms*—** Knowledge distillation, edge intelligence, data-free, attention, ensemble

1. INTRODUCTION

Recent deep learning-based edge intelligence applications, such as the medical ones on wearable devices and immersive video analysis, require tiny model and agile model update strategy. However, current models need hundreds of millions of bits for storage. For example, the size of ResNet-152 is 232MB. In order to deploy these applications on edge devices (e.g., smartbands and smart helmets) with constrained computational and storage resources, the size and complexity of models should be as small as possible. Therefore, model compression, which can generate a tiny model with existing trained models, is desirable for edge intelligence applications.

Current methods for compressing a trained neural network into a tiny one can be categorized into quantization, pruning and KD. Quantization compresses the trained neural network by representing each weight with fewer bits. Pruning explores the redundancy in model weights, and removes redundant and noncritical ones [1]. Both neural network quantization and pruning only exploit information from a single target model, while KD is able to utilize information from other models. This paper is based on KD, where a large-size teacher model is utilized to guide the learning of a compressed student model. KD often needs the original data (for training teacher model) to train the student. However, the original data are usually unavailable due to the privacy issue or huge transmission burden. In order to conduct KD with minimum number of original samples, Li *et al.* [2] proposed to prune the trained teacher model to obtain the initial student model first, and then conduct KD by fitting the block-level outputs from the student to the teacher. The block-level outputs fitting was supervised by few samples. Chen *et al.* [3] developed a data-free KD method that uses a generator to synthesize data as the alternative of the original data. A major drawback of these methods is that a single teacher model is utilized, and hence the learned student model may be not reliable due to the limited number of original data.

To improve the reliability, a group-stack dual-GAN structure is proposed for data-free knowledge amalgamation with multiple teachers [4]. In this structure, a generator is designed to synthesize data that are used to mimic the original data, and a dual-generator is designed to amalgamate knowledge from multiple teachers by taking the output from the former generator as input. However, amalgamating knowledge from multiple teachers by using a single model ignores the intrinsic diversity of different teachers, and this may lead to sub-optimal performance in the supervision of student model.

To remedy these drawbacks, we design a collaborative data-free knowledge distillation (CDFKD) framework to preserve the diversity of multiple teachers in the training of a tiny student and explore the teacher contributions in the prediction. In particular, we first design a multi-teacher data generation module that leverages generative adversarial network (GAN) [5] to synthesize data, where an ensemble of multiple teachers is adopted as the discriminator. In this way, the generator is able to synthesize diverse types of data. Then the synthetic data are utilized for the subsequent multi-teacher

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KD, where a multiple headers module is incorporated to simultaneously exploit knowledge from multiple teachers for robust distillation, and the diversity of multiple teachers can be carefully preserved by the multi-header structure. Finally, when few original samples are available, an enhanced attention module is added to adaptively aggregate the outputs predicted by the multiple headers of student model.

The proposed framework is trained by a two-stage strategy and the main contributions of this paper are:

- We propose a novel data-free KD-based model compression framework, where multiple teachers are cooperated for robust KD;
- We design both novel multi-teacher data generation and KD modules that can effectively exploit diversity of different teachers, and this is the first work that employs a multi-header structure in data-free distillation;
- We develop an adaptive aggregation module to further improve the performance when only a few original data are available.

We conduct extensive experiments on MNIST dataset and CIFAR-10 dataset. Results demonstrate that the proposed method achieves significant accuracy improvements on the two datasets (4.48% and 2.96%, respectively) compared with the data-free model compression counterpart. Some carefully designed ablation studies are also performed to verify the effectiveness of different components in our framework.

2. RALATED WORK

KD is a common model compression method. In this section, we summarize KD methods for both data-free and multi-teacher scenarios.

Data-free Distillation Data-free distillation transfers knowledge from a trained teacher model to a tiny student model without using the original training set. Most of existing data-free KD methods employ synthetic data to train the student. For example, Lopes *et al.* [6] collected metadata during training the teacher with the original data, and then reconstructed the origin data with help of the metadata to train the student. Nayak *et al.* [7] directly synthesized "Data Impressions", and then used the "Impressions" to mimic the original data. Chen *et al.* [3] introduced a generator to synthesize data, where the trained teacher was used as a discriminator to guide the image generation procedure. Ye *et al.* [4] proposed a group-stack dual-GAN structure, where a generator was proposed to synthesize images, and a dual-generator was designed to produce label vectors. All of these methods only adopt a single teacher for student supervision, and hence the learned student model may be not reliable with no or only a few original data.

Multi-teacher Distillation A student can access more than one teacher in multi-teacher distillation scenario. The complementary knowledge from multiple teachers can improve the performance of the student. Hinton *et al.* [8] proposed that knowledge from multiple teachers can be distilled by simply utilizing an ensemble of teachers for supervision. You *et al.* [9] trained a tiny student model by averaging soft outputs from teachers at the output layer. They also enforced teachers and the student to be similar at intermediate layers. Zhang *et al.* [10] proposed to train multiple students collaboratively, and enable students to teach each other in the training. All of these approaches require the original data for distillation, while such data may be unavailable due to the privacy issue and huge transmission burden.

3. FRAMEWORK DESIGN

In this section, we depict the proposed framework for collaborative data-free KD.

3.1. Overview

We design a neural network architecture to transfer knowledge from the ensemble of multiple teachers to a multi-header student without using the original samples for training. Generally speaking, given a set of trained teachers and a multi-header student, our goal is to teach the student to mimic the behavior of the ensemble of teachers.

Our proposed architecture is illustrated in Figure 1. There are mainly three components: multi-teacher data generation, multi-teacher KD, and adaptive outputs aggregation. In the first component, a generator is trained to take random noise as input, and then output synthetic data (such as images for illustration) under the supervision of the ensemble of teachers. Such data will be passed to the second component and labeled by the ensemble of teachers. In the second component, the multi-header student will be trained using synthetic data with soft labels. Each header of the student captures the behavior of a corresponding teacher. In the third component, we introduce an attention mechanism to adaptively aggregate the multiple outputs predicted by the multi-header student. The attention module is optional in our method, and is activated when few original data are available.

The proposed model is trained by a two-stage strategy. In the first stage, the student and the generator are trained alternatively in each epoch. In the second stage, we fix weights of the student, and train the attention network with only a few original data. In the prediction, the aggregated output is regarded as the final result. More details of each component are described as follows.

3.2. Multi-teacher data generation

In this component, we synthesize data using a GAN. In the proposed structure, a generator receives random noise and synthesizes data, where multiple teachers are integrated to discriminate samples that are similar to the original data.

A common GAN is composed of a generator G and a discriminator D, both of which are neural networks. Given a prior $p_{\mathbf{z}}(\mathbf{z})$ on input noise variables $\mathbf{z}$, G maps $\mathbf{z}$ to the generated data $G(\mathbf{z})$, and then $G(\mathbf{z})$ and real samples are mixed to train D. D is designed to differentiate training examples

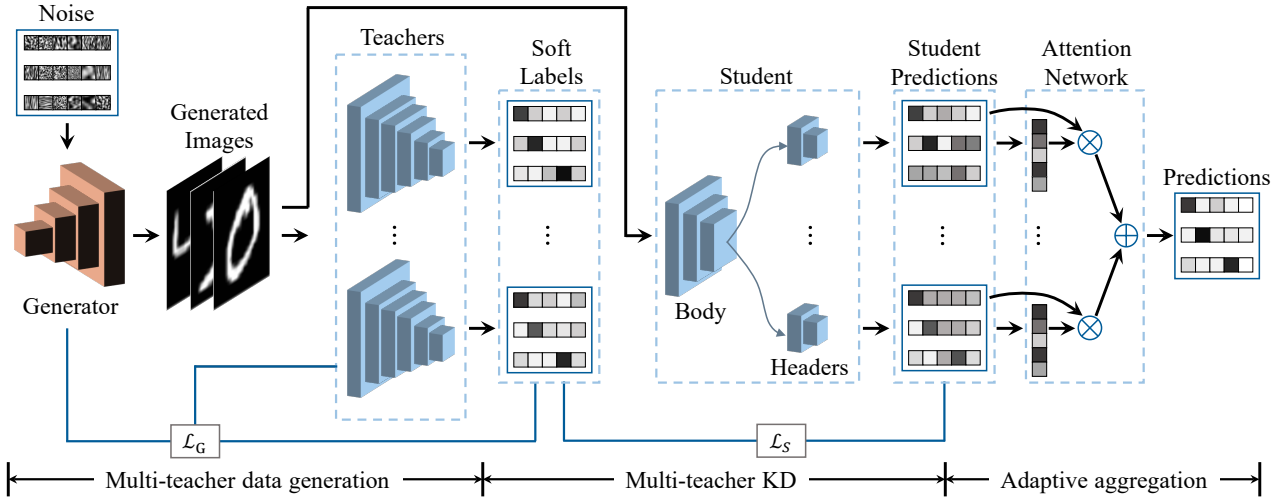

Fig. 1. **Network Architecture** of the proposed CDFKD framework. Firstly, the generator takes noise vectors as input and output synthetic data (such as images for illustration), where an ensemble of multiple teachers is utilized for discrimination. Then a student with multiple headers is trained using the generated data and corresponding soft labels produced by the ensemble of teachers. Finally, outputs of the multi-header student are adaptively aggregated using an enhanced attention module for final prediction.

and synthetic data. The GAN is optimized with the objective function $V(G, D)$:

$$\min_G \max_D V(G, D) = \mathbb{E}_{\mathbf{x}\sim p_{data}(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]. \tag{1}$$

However, training of the discriminator requires the original data. We cannot optimize a GAN without the original dataset.

In our method, the ensemble of teachers is regarded as the discriminator. Recent work [3] has shown that the trained teacher can perform as a discriminator for image generating task with a carefully designed objective function, since the teacher trained on a large-scale dataset can extract semantic features from images. Trained teachers are denoted as $\{\mathcal{T}_1, \mathcal{T}_2, \cdots, \mathcal{T}_M\}$. For a batch of synthetic images $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\}$, $\mathcal{T}_i(\mathbf{x}_j)$ represents the distribution of the j-th image $\mathbf{x}_j$ in the batch, which is predicted by the i-th teacher $\mathcal{T}_i$. Then we define the pseudo ground-truth label of $\mathcal{T}_i(\mathbf{x}_j)$ as $\mathbf{t}_i^j = \arg\max_k(\mathcal{T}_i(\mathbf{x}_j))_k$, where k is the index of the vector $\mathcal{T}_i(\mathbf{x}_j)$. Objective functions of this phase are:

$$\begin{aligned} \mathcal{L}_{oh} &= \frac{1}{MN}\sum_{i,j}\mathcal{H}_{cross}(\mathcal{T}_i(\mathbf{x}_j), \mathbf{t}_i^j), \\ \mathcal{L}_a &= -\frac{1}{MN}\sum_{i,j}||\mathbf{f}_i^j||_1, \\ \mathcal{L}_{ie} &= -\frac{1}{M}\sum_i \mathcal{H}_{info}(\sum_j \mathcal{T}_i(\mathbf{x}_j)), \\ \mathcal{L}_G &= \mathcal{L}_{oh} + \alpha\mathcal{L}_a + \beta\mathcal{L}_{ie}, \end{aligned} \tag{2}$$

where $\mathcal{L}_{oh}$ forces teachers to output one-hot like codes, $\mathcal{L}_a$ encourages higher activations from the last convolution layer of teachers, $\mathcal{L}_{ie}$ balances the number of samples in each class within a mini-batch, and $\mathcal{L}_G$ merges aforementioned terms. In $\mathcal{L}_{oh}$, $\mathcal{H}_{cross}$ is a cross entropy loss. In $\mathcal{L}_a$, $\mathbf{f}_i^j$ is the feature map of $\mathbf{x}_j$ extracted by the last convolution layer of $\mathcal{T}_i$. In $\mathcal{L}_{ie}$, $\mathcal{H}_{info}$ is an information entropy loss. In $\mathcal{L}_G$, α and β are hyper parameters balancing different terms. The design of the data generation module refers to the structure in [3], and we make some modifications for multi-teacher scenario.

Discussion. We introduce a generator to synthesize data for training, and the generated data will be passed to the second component, so that we can achieve KD without using the original data.

3.3. Multi-teacher knowledge distillation

In this component, we transfer knowledge from the ensemble of teachers to a single student with help of the generated data. The student contains a multiple output headers structure to preserve diverse knowledge of different teachers.

Complementary knowledge may benefit the performance of the ensemble of models. If we train the student with the average of soft labels, the complementarity of multiple teachers will vanish. To tackle this problem, we replace the last several layers of the student with a multi-header output structure. Each header in this structure is simply a copy of the replaced layers. To preserve the complementary knowledge within a single student with multiple headers, each header is assigned to a corresponding teacher. Thus the number of teachers and the number of headers are equal. During training, each header should mimic the behavior of its corresponding teacher [11]. We introduce Kullback-Leibler divergence to measure the similarity between the outputs from headers and teachers. The student is denoted as $\mathcal{S}$. Given a synthetic

image $\mathbf{x}$, $\mathcal{S}_i(\mathbf{x})$ is the distribution of $\mathbf{x}$ predicted by the i-th header of the student. The loss function is defined as:

$$\mathcal{L}_{headers} = \frac{1}{M}\sum_i \mathcal{H}_{kd}(\mathcal{T}_i(\mathbf{x})||\mathcal{S}_i(\mathbf{x})), \tag{3}$$

where $\mathcal{H}_{kd}(p||q) = p\ln p - p\ln q$. Baxter [12] noted that the hard parameters sharing structure at the front of student could reduce the probability of over fitting. By distilling knowledge from the ensemble of teachers into a single model, the fitting bias of different teachers will be averaged automatically. Thus the complementary knowledge will help the multi-header student outperform a single-header student.

It is reasonable that the student should mimic the behavior of the ensemble of teachers, since ensemble of models usually performs better than a single model. Thus another KL divergence loss term is used to reduce the dissimilarity between the ensemble of teachers and the ensemble of headers:

$$\mathcal{L}_{ensemble} = \mathcal{H}_{kd}(\hat{\mathcal{T}}(\mathbf{x})||\hat{\mathcal{S}}(\mathbf{x})), \tag{4}$$

where $\hat{\mathcal{T}}$ and $\hat{\mathcal{S}}$ are the ensemble of teachers and headers, respectively. The ensemble is achieved by simply averaging output logits. Different models usually make different mistakes on the test set [13]. By imitating the behavior of the ensemble of teachers, mistakes made by a single header will be corrected by other headers.

Finally, the student is trained by:

$$\mathcal{L}_S = \mathcal{L}_{headers} + \gamma\mathcal{L}_{ensemble}, \tag{5}$$

where γ balances the two terms in the objective function. During distilling, $\mathcal{L}_{headers}$ will prompt each header to learn from its corresponding teacher. $\mathcal{L}_{ensemble}$ will prompt the ensemble of headers to imitate the ensemble of teachers, which guides the distillation procedure from the global perspective.

Discussion. We design a student with multiple output headers for data-free collaborative KD. By training the student with the loss function (5), the diversity of teachers will be preserved by multiple headers, which will improve the performance of the student.

3.4. Adaptive outputs aggregation

In this component, we design an enhanced attention network to further improve the performance of our method. When few data are available, the attention network can weightedly and adaptively aggregate the outputs of different headers rather than simply averaging them.

The diversity has been preserved by the multi-header student. With few original data, the attention module will exploit the preserved diversity to further improve the performance of the student. In the proposed architecture, a fully connected layer is applied to compute attention weights. When the trained student receives an original image, the fully connected layer reads output logits from all headers, and computes weights for them. These weights are then passed to a softmax layer to get the normalized weight e_i, where $\sum_i e_i = 1$. Let $\mathbf{s}_i$ be the output logits vector from the i-th header, $\mathbf{q}$ be parameters of the fully connected layer, the attention mechanism can be described as follows:

$$e_i = \frac{\exp(\mathbf{q}^T\mathbf{s}_i)}{\sum_j \exp(\mathbf{q}^T\mathbf{s}_j)}, \tag{6}$$

$$\mathbf{s}' = \sum_i e_i\mathbf{s}_i. \tag{7}$$

The attention network is trained by few original data with cross-entropy loss.

Existing methods for KD without the whole original dataset work with either few data or no data. In practice, whether we can acquire few data depends on many complicated factors (e.g., privacy or transmission consumption). Thus previous methods may be unusable in mismatched scenario. In our method, we first train the student without any original data. Once we can acquire few original data, the single-layer attention network will improve the performance of the student by aggregating outputs from headers in a more reasonable way. The attention module improves the flexibility of our method by extending its scope to few-data scenario. Because of the simple architecture of the attention network and the compact outputs from headers, the training process of the attention network only takes several minutes.

Discussion. We introduce an enhanced attention mechanism to aggregate the outputs from the student. The enhanced module improves the flexibility of our method. When few original data are available, the module will exploit the diversity of headers and generate better predictions.

4. EXPERIMENTS

In this section, we evaluate the performance of the proposed collaborative data-free KD method on MNIST [14] and CIFAR-10 [15] datasets. We also design ablation experiments to explore the effectiveness of each component in the model. We conduct 5 trials for each experiment setting.

4.1. Experiments on MNIST

We first implement experiments on MNIST. MNIST is a dataset of handwritten digits, which contains 60000 training samples and 10000 testing samples. The dataset is composed of images from 10 categories. The size of each image is 28×28 pixels. We randomly sample 1% data from the training set, and regard these data as the few original samples that we can acquire in practice. We evaluate the performance of our proposed method on the whole test set.

We train 3 models as teachers, each of them is a 5-layer CNN. The first 3 layers are convolution layers and the last two layers are fully connected layers. Numbers of channels (units) in each layer are 2-4-60-30-10, respectively. The structure of the student is similar to the teacher. At each layer, the number of channels (units) in each layer of the student is half of the number of channels (units) in each layer of the teacher.

Table 1. Classification result on MNIST dataset

Algorithm	Required data	FLOPs	#Params	Accuracy(%)
Ensemble teacher	Original Data	~ 208K	~ 25.0K	98.28
KD [8]	Original Data	~ 28K	~ 2.2K	97.45 ± 0.20
Hydra [11]	Original Data	~ 29K	~ 3.5K	97.47 ± 0.13
DAFL [3]	No Data	~ 28K	~ 2.2K	89.47 ± 5.61
Ours(without attention)	No Data	~ 29K	~ 3.5K	93.57 ± 1.12
Ours	1% Original Data	~ 29K	~ 3.5K	$\mathbf{93.95 \pm 0.99}$

We regard the last two fully connected layers as the header. Compared with the ensemble of teachers, our model contains significantly fewer parameters. The structure of the generator follows that in [3]. We determine the value of each hyperparameter by grid search. α, β and γ are 0.1, 5 and 1 respectively. In the first stage, the generator and multi-header student are trained for 200 epochs with Adam optimizer, and the learning rate are 0.2 and 0.002, respectively. In the second stage, the attention module is trained on data sampled from the training set for 100 epochs with Adam optimizer with the learning rate being 0.01.

Table 1 reports results of different methods on the MNIST dataset. Trained with the original training set, the ensemble of teachers achieves 97.67% accuracy. We first test two full-data KD methods designed for multiple teachers. Method of Hinton [8] achieves 97.45% accuracy on average, which regards the ensemble of teachers as a single model. Hydra [11] introduced a multi-header student, and prompts each header to learn from its corresponding teacher. The accuracy of Hydra is 97.47% on average.

We then implement KD in the data-free scenario. In each trial of an experiment setting, we utilized a generator with the same randomly initialized parameters to eliminate the influence of the initialization. DAFL achieves 89.47% accuracy on average with a high variance by synthesizing images and transferring knowledge of a single teacher to the student. The proposed method achieves 93.57% accuracy on average after finishing the training of the first stage, where final predictions are computed by averaging the outputs from headers coarsely. In the second stage, we aggregate the outputs from headers by the attention network trained with sampled training data. The final accuracy of our method achieves 93.95% on average.

4.2. Experiments on CIFAR-10

We conduct experiments on CIFAR-10 dataset to further explore the effectiveness of our method. CIFAR-10 is a dataset that involves 50000 training samples and 10000 testing samples. Few original data are simulated by sampling 1% data from the training set.

Due to the memory limitation of GPU, we employ modified ResNet-34 and ResNet-10 as the teacher and the student, respectively. The modification is implemented by reducing channels of each layer to a quarter. We still train 3 models as teachers. The generator is trained with Adam optimizer, whose learning rate is 0.001. The student is optimized by stochastic gradient descent (SGD) optimizer with learning rate 0.1, which is divided by 10 at epoch 400 and 600. The generator and student are trained alternatively for 800 epochs.

Table 2 reports experiment results on CIFAR-10 dataset. The ensemble of teachers achieves 94.43% accuracy on average. By distilling knowledge from the ensemble of teachers with the original training set, the method of Hinton [8] achieves 92.34% accuracy and Hydra [11] achieves 92.58% accuracy on average.

Results of data-free KD are similar to results on MNIST. DAFL [3] achieves 86.89% accuracy on average. With a multi-header structure, our method can achieve 89.71% accuracy on average while keeping a lower variance than DAFL. After training the attention module with the sampled data, our method finally achieves 89.85% accuracy on average.

4.3. Ablation experiments

Ablation experiments are conducted on CIFAR-10 dataset to verify the effectiveness of the multi-header structure and each loss term in Equation (5). Details of ablation experiments are introduced in this subsection.

The settings of ablation experiments are the same as those in Section 4.2. Results of experiments are reported in Table 3. We utilize the student without multiple headers to learn from the ensemble of teachers. The implementation of the distillation follows that in [8], where the synthetic data replaces the original data. The student achieves 88.86% accuracy on average. When the multi-header structure is introduced, we train the student with $\mathcal{L}_{headers}$ alone. The student achieves 89.16% accuracy after the training of the first stage. Then we train the attention network. Finally the student achieves 98.36% accuracy on average. We further leave $\mathcal{L}_{ensemble}$ alone to train the multi-header student. The student can achieve 89.14% accuracy without the attention module and 89.40% accuracy after finishing the training of the attention network. By contrast, the student with multiple headers, attention module, and all loss terms achieves the best performance, which is 89.85% accuracy on average.

The ablation experiments demonstrate the effectiveness of each component in our proposed architecture. $\mathcal{L}_{headers}$ works together with multiple headers to capture the complementary knowledge from different teachers, while $\mathcal{L}_{ensemble}$ guides the distillation from the global perspective. The attention network helps to aggregate the complementary outputs in a more reasonable way than simply averaging them.

Table 2. Classification result on CIFAR-10 dataset

Algorithm	Required data	FLOPs	#Params	Accuracy(%)
Ensemble of teachers	Original Data	~ 220M	~ 4.0M	94.43
KD [8]	Original Data	~ 35M	~ 0.7M	92.34 ± 0.09
Hydra [11]	Original Data	~ 52M	~ 1.8M	92.58 ± 0.12
DAFL [3]	No Data	~ 35M	~ 0.7M	86.89 ± 0.98
Ours(without attention)	No Data	~ 52M	~ 1.8M	89.71 ± 0.20
Ours	1% Original Data	~ 52M	~ 1.8M	$\mathbf{89.85 \pm 0.18}$

Table 3. Ablation experiments result on CIFAR-10

Algorithm	Accuracy(%)
Single-header structure	88.86 ± 0.16
$\mathcal{L}_{headers}$	89.16 ± 0.19
+ attention	89.36 ± 0.15
$\mathcal{L}_{ensemble}$	89.14 ± 0.47
+ attention	89.40 ± 0.47
$\mathcal{L}_{ensemble} + \mathcal{L}_{headers}$	89.71 ± 0.20
+ attention	$\mathbf{89.85 \pm 0.18}$

5. CONCLUSION

In this paper, we proposed a three-stage data-free collaborative KD framework, where multi-teacher generator and multi-header module are designed to exploit the diversity of different teachers. When few original data are available, an enhanced attention module is developed to further improve the performance by exploring the significance of different teachers. Effectiveness of the proposed framework is verified on two popular datasets. From the results, we mainly conclude that: 1) it is beneficial to integrate multiple teachers for data-free KD, but a moderate number of parameters should be added; 2) all the multi-teacher ensemble, multi-header distillation and adaptive aggregation are critical in performance improvement. In the future, we intend to find some strategies to select the most important data for model training and incorporate some pruning or quantization techniques to reduce the number of parameters.

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